

 ENTERPRISE NETWORK SECURITY LAB

VERTEX Network Security

Implementation of Enterprise Network Security & Virtualization Infrastructure
Practical Cyber Attack Simulations & Advanced Defense Mechanisms via OPNsense Firewall.

Project Overview

- ▶ **Enterprise Virtual Lab:** Built to simulate realistic cyber attack vectors and deploy effective boundary defense mechanisms.
- ▶ **Network Behavior Analysis:** Comparative evaluation measuring network performance and security posture before and after control enforcement.
- ▶ **Core Focus Areas:** Enforcing granular firewall rules, strict VLAN network segmentation, and real-time analytical logging.
- ▶ **Structured Repository:** Includes 6 fully documented attack scenarios backed by live Proof-of-Concept (PoC) evidence.



Core Objectives



Attack Analysis

Comprehend execution mechanics of network and web attacks and quantify their impact on enterprise infrastructure.



Defense Deployment

Implement multi-layered defenses utilizing OPNsense firewall paired with host-level security controls.



Network Isolation

Enforce strict DMZ and LAN network segmentation to block unauthorized lateral movement.



Validation & Evidence

Measure operational metrics comparing baseline vs hardened states with documented PoC logs.

Lab Architecture

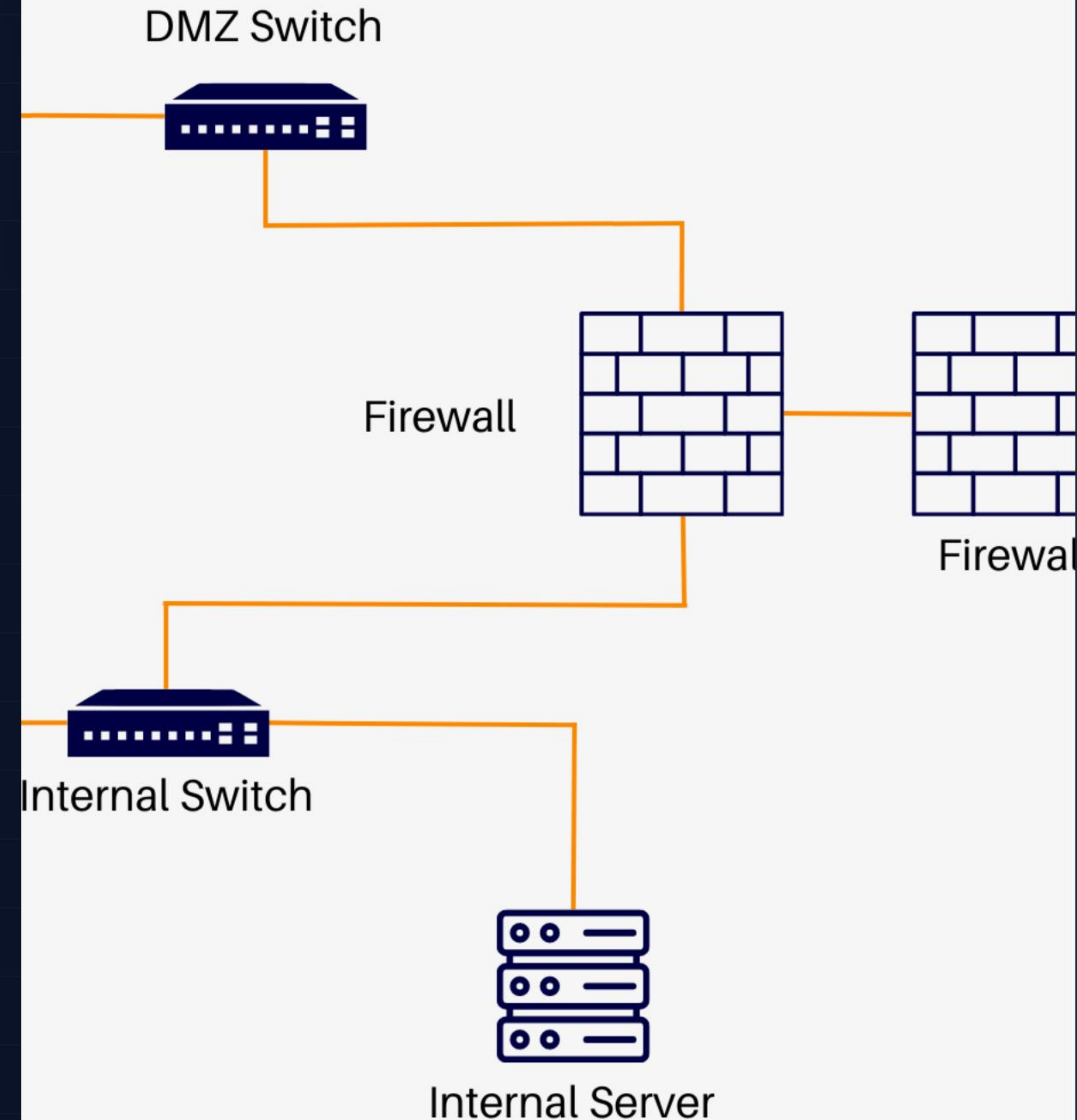
Virtual Infrastructure Layout

Attacker Station (Kali Linux): Dedicated workstation tailored for executing penetration testing and exploit scenarios.

OPNsense Firewall: Central gatekeeper enforcing routing, NAT, traffic filtering, and intrusion prevention policies.

DMZ & LAN Zones: External DMZ hosting Web Services and internal LAN containing Windows Enterprise endpoints.

Target Vulnerable Host: Metasploitable 2 instance configured for controlled security testing.



Attack Scenarios Matrix

Attack Vector	Demonstrated Threat / Impact	Primary Defense Control
Brute Force	Iterative automated credential guessing attempts	Account Lockout + Strong Auth + Rate Limiting
Lateral Movement	Pivoting from public DMZ into private internal LAN	VLAN Segmentation + Unidirectional Firewall Rules
DoS / DDoS	Service disruption via high-volume traffic flooding	Rate Limiting + OPNsense Syncookies + Inspection
MITM / ARP Spoofing	Data interception via ARP cache table poisoning	Static ARP + Dynamic ARP Inspection + Encryption
Nmap Reconnaissance	Port scanning and host service discovery	Port Hardening + Firewall Filter Stealth Block
SQL Injection & XSS	Exploiting web input parameters & script execution	Parameterized Queries + Input Sanitization + Encoding

OPNsense Security Layer

Centralized Traffic Filtering

Gatekeeper Architecture: OPNsense operates as the core perimeter firewall controlling packet flow between Attacker, DMZ, and LAN zones.

Default-Deny Policy: Blocks all inter-zone communication by default, allowing only explicitly defined and authorized network traffic.

Granular Control & Telemetry

Fine-Grained Policies: Rules configured based on precise combinations of Source IP, Destination IP, Protocol type, and Port numbers.

Audit Logs & Proof: Real-time firewall logging captures active rule triggers to empirically verify mitigation effectiveness.

Before vs. After Security

Unprotected State

- ▶ Direct access for attack traffic to reach target hosts.
- ▶ Unrestricted public exposure of administrative ports.
- ▶ Unchecked lateral pivot capability from DMZ to LAN.
- ▶ Severe service degradation during flood attacks.

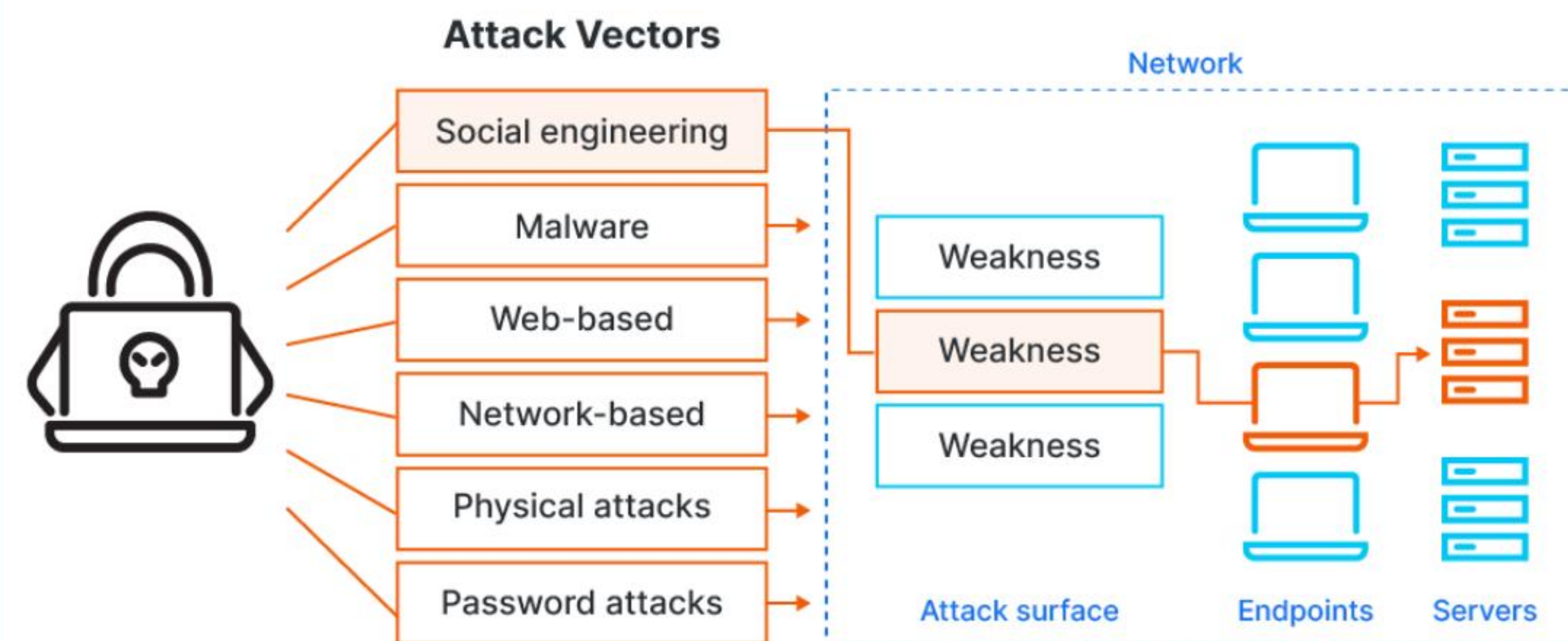
Protected State

- ▶ Unauthorized traffic blocked or rate-limited at perimeter.
- ▶ Unnecessary ports closed and services hardened.
- ▶ Strict Zero-Trust isolation preventing DMZ to LAN pivoting.
- ▶ Live log evidence confirming successful mitigation.

Brute Force Defense

Mitigation Strategy & Implementation

- ▶ **Attack Phase:** Simulating rapid automated login attempts against network services within the virtual lab environment.
- ▶ **Defense Phase:** Configuring Account Lockout policies upon threshold breaches, mandatory password complexity, and connection rate limiting.
- ▶ **Evidence & Logs:** Captured evidence and analytical logs cataloged under /Attacks/Brute_Force directory.



DMZ & LAN Isolation

Preventing Lateral Movement

Core Objective: Ensure that a compromised public web server in the DMZ cannot serve as a stepping stone into the internal enterprise LAN.

Strict Rules: Firewall policies mandate unidirectional isolation, dropping all network connection requests originating from DMZ to LAN.

Validation & Verification

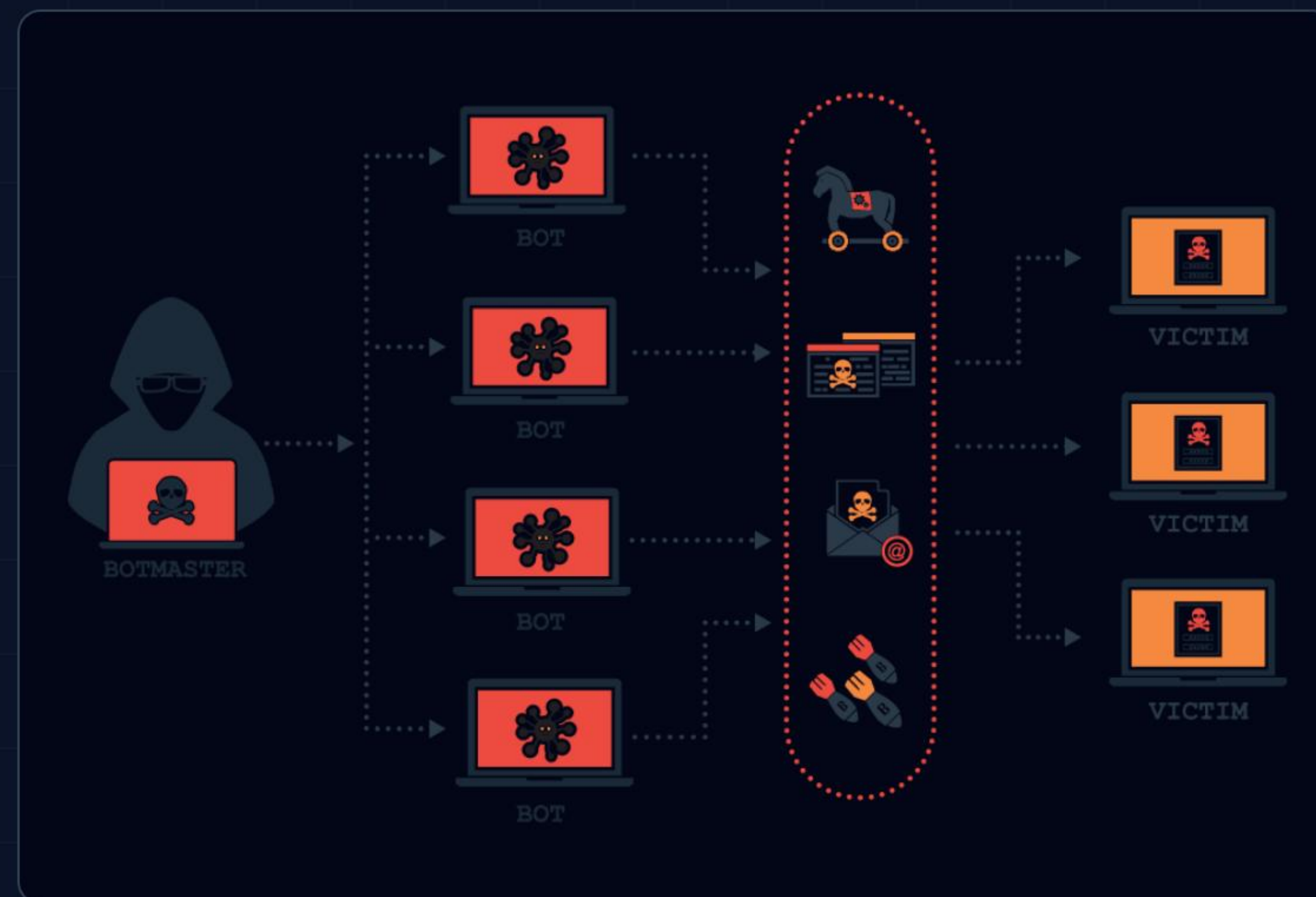
Connectivity Sweeps: ICMP and TCP probe attempts executed from DMZ endpoints towards internal workstations.

Proven Defense: Audit logs confirm 100% block rate of unauthorized connection attempts post-rule activation.

DoS & DDoS Defense

Managing High-Volume Traffic

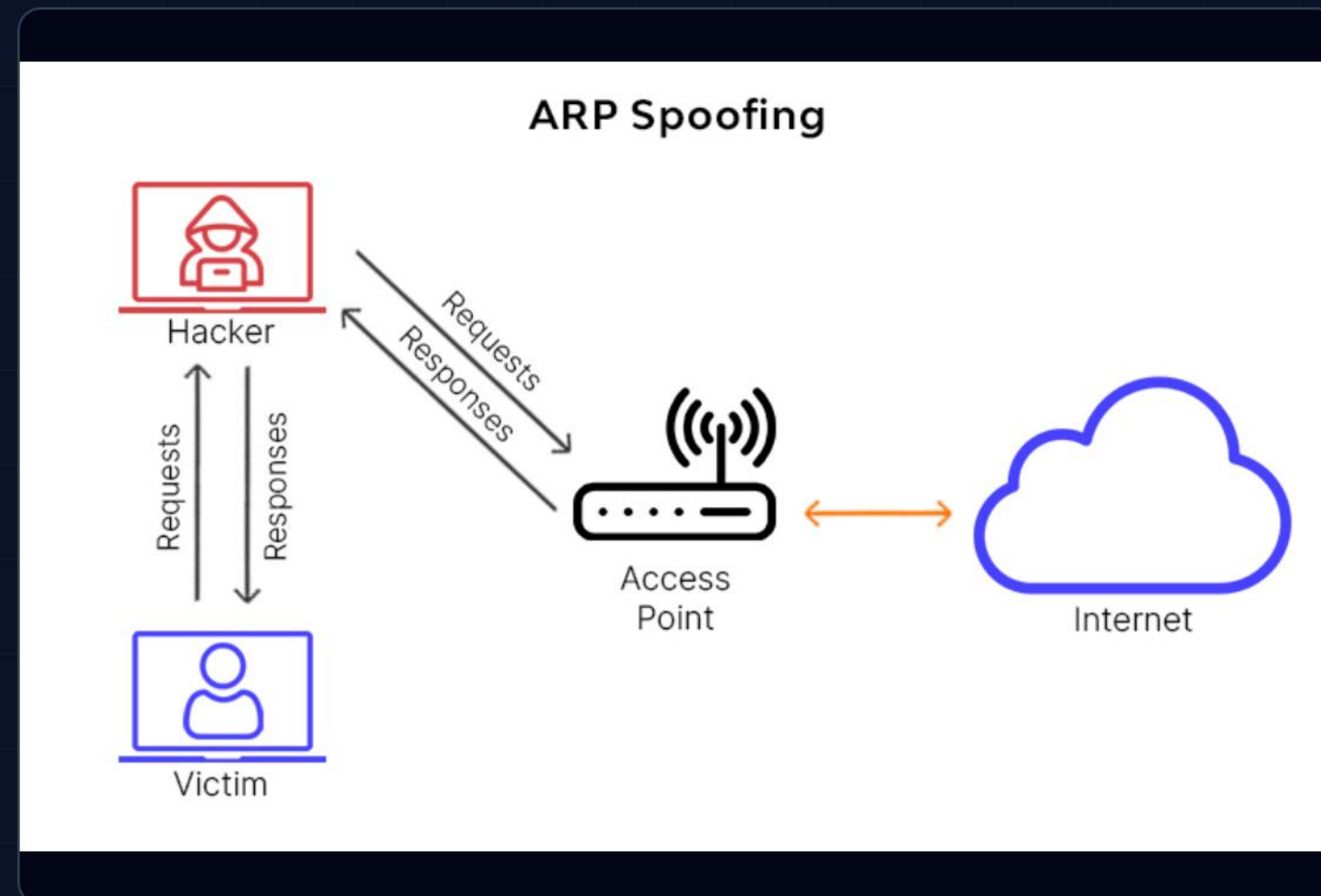
- ▶ **Controlled Flood Simulation:** Generating massive synthetic traffic bursts inside the lab to test system resilience.
- ▶ **Mitigation Strategy:** Enforcing dynamic connection rate limits, OPNsense SYN-flood protection, and traffic shaping rules.
- ▶ **Resource Monitoring:** Telemetry tracking CPU, memory consumption, and packet loss ratios during attack runs.



MITM & ARP Spoofing

Man-in-the-Middle Countermeasures

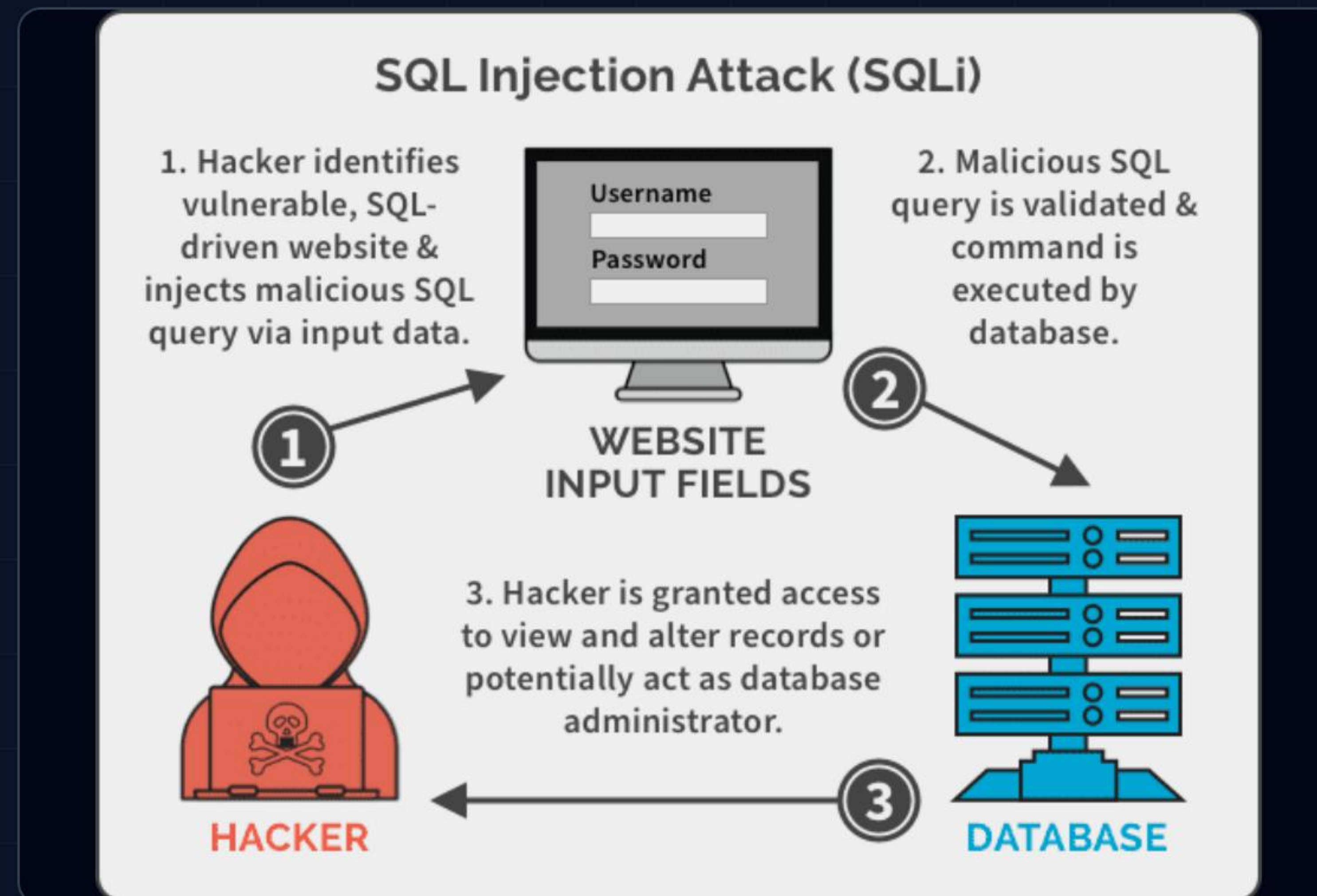
- ▶ **Threat Model:** Attacker manipulating local ARP tables to intercept confidential host-to-gateway traffic.
- ▶ **Defensive Controls:** Implementation of Static ARP bindings, Dynamic ARP Inspection (DAI), and mandatory TLS encryption.
- ▶ **Active Auditing:** Continuous network monitoring to flag unexpected ARP cache modifications or broadcast anomalies.



Web & Recon Attacks

Application & Scanning Protection

- ▶ **Nmap Recon Defense:** Closing unused listening ports and dropping stealth probe signatures at firewall boundary.
- ▶ **SQL Injection Mitigation:** Implementing parameterized prepared queries and strict input validation schemas.
- ▶ **XSS Safeguards:** Enforcing Context-Aware Output Encoding and strict Content Security Policy (CSP) rules.



Security Impact Metrics

100%

Blast Radius Containment

Measurable Outcomes

Defense-in-Depth Architecture: Combining host-level hardening with boundary firewall controls guarantees comprehensive protection.

Minimized Attack Surface: Strict port access restrictions limit exposed services strictly to necessary functions.

Verifiable Security Posture: Empirical log analysis provides solid Proof-of-Concept validation for enterprise readiness.

Questions & Discussion

Thank you for reviewing the VERTEX Network Security Implementation.

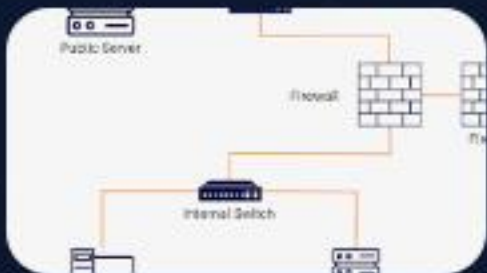
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Image Sources



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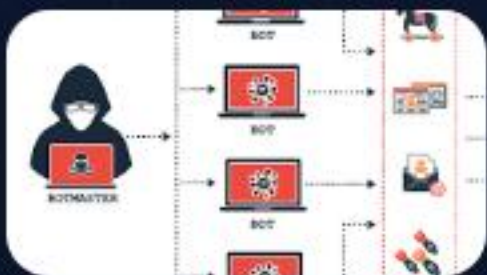
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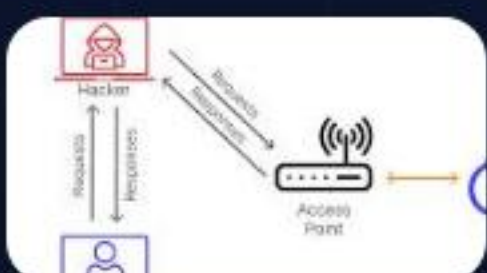
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